1. forward: $\begin{cases} \text{for every kernel} \Rightarrow z = w * x \\ \text{for output} \Rightarrow y = \sum_{i=1}^{c} z_i = \sum_{i=1}^{c} w_i * x_i \ , \ i = 1,2,\dots,c \\ \text{for position } (i,j) \Rightarrow y_{i,j} = z_{i,j,1} + \dots z_{i,j,c} \end{cases}$

given that $\frac{\partial J}{\partial y} = G$ and $\frac{\partial y}{\partial z_i} = 1$ and $\frac{\partial z_{i,j,l}}{\partial x_{p,q,l}} = w_{p-i,q-j,l}$

for certain channel $l \Rightarrow \frac{\partial J}{\partial x_l} = rot_{180^\circ}(w_l)$

Therefore, $\frac{\partial J}{\partial x} = stack_{channel}\{rot_{180^\circ}(w_1) * G, \dots, rot_{180^\circ}(w_c) * G\} \in R^{h \times w \times c}$

2. first convolution $\Rightarrow z_{i,j} = \sum_{a,b} w^1_{a,b} x_{i+a,j+b}$, $a,b \in \{0,\dots,k_1-1\}$
second convolution $\Rightarrow y_{i,j} = \sum_{u,v} w^2_{u,v} z_{i+u,j+v}$, $u,v \in \{0,\dots,k_2-1\}$

$\Rightarrow y_{i,j} = \sum_{u,v} w^2_{u,v} \left(\sum_{a,b} w^1_{a,b} x_{i+u+a,j+v+b} \right) = \sum_{u,v} \sum_{a,b} w^2_{u,v} w^1_{a,b} x_{i+(u+a),j+(v+b)}$

let $p = u+a$, $q = v+b$, $p,q \in \{0,\dots,k_1+k_2-2\}$

$\Rightarrow y_{i,j} = \sum_{p,q} \left(\sum_{u,v} w^2_{u,v} w^1_{p-u,q-v} \right) x_{i+p,j+q}$

let $w^3_{p,q} = \sum_{u,v} w^2_{u,v} w^1_{p-u,q-v}$

$\Rightarrow y_{i,j} = \sum_{p,q} w^3_{p,q} x_{i+p,j+q} \Rightarrow y = w^3 * x$

$\Rightarrow w^1(k_1) + w^2(k_1) \rightleftharpoons w^3(k_1+k_2-1)$

3. $\mu = \frac{1}{N} \sum_{i=1}^{N} x^{(i)}$ $\sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x^{(i)} - \mu)^2$ $s = \sqrt{\sigma^2 + \varepsilon}$

$\hat{x}^{(i)} = \frac{x^{(i)} - \mu}{s}$ $y^{(i)} = a\hat{x}^{(i)} + \beta$ $g^{(i)} = \frac{\partial J}{\partial y^{(i)}}$

$\Rightarrow \frac{\partial J}{\partial a} = \sum_{i=1}^{N} g_i x^{(i)}$ $\frac{\partial J}{\partial \beta} = \sum_{i=1}^{N} g^{(i)}$

$\because \sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x^{(i)2} - 2x^{(i)}\beta + \beta^2) = \frac{1}{N} \sum_{i=1}^{N} x^{(i)2} - 2\beta \frac{1}{N} \sum_{i=1}^{N} x^{(i)} + \beta^2 = \frac{1}{N} \sum_{i=1}^{N} (x^{(i)2} - \mu^2)$

$\therefore \frac{\partial \sigma^2}{\partial x^{(i)}} = \frac{2}{N} x^{(i)} - 2\mu \frac{\partial \mu}{\partial x^{(i)}} = \frac{2}{N} x^{(i)} - \frac{2}{N} \mu = \frac{2}{N} (x^{(i)} - \mu)$

$\therefore \frac{\partial \hat{x}^{(m)}}{\partial x^{(i)}} = \frac{\partial (x^{(m)} - \mu)}{\partial x^{(i)}} \cdot \frac{1}{N\sqrt{\sigma^2+\varepsilon}} + \frac{\partial \frac{1}{N\sqrt{\sigma^2+\varepsilon}}}{\partial x^{(i)}} \cdot (x^{(m)} - \mu)$

$= (\delta_{mi} - \frac{1}{N}) \cdot \frac{1}{N\sqrt{\sigma^2+\varepsilon}} + (-\frac{1}{2})(\sigma^2+\varepsilon)^{-\frac{3}{2}} \cdot \frac{2}{N}(x^{(i)} - \mu)$

$= \frac{1}{s}(\delta_{mi} - \frac{1}{N}) - \frac{(x^{(m)} - \mu)(x^{(i)} - \mu)}{Ns^3}$

$= \frac{1}{s}(\delta_{mi} - \frac{1}{N} - \frac{\hat{x}^{(m)} \hat{x}^{(i)}}{N})$ 且 $\delta_{mi} = \begin{cases} 1, & m = i \\ 0, & m \neq i \end{cases}$

$\therefore \frac{\partial J}{\partial x^{(i)}} = \sum_{m=1}^{N} \frac{\partial J}{\partial \hat{x}^{(m)}} \frac{\partial \hat{x}^{(m)}}{\partial x^{(i)}} = \sum_{m=1}^{N} a g^{(m)} \frac{1}{s} (\delta_{mi} - \frac{1}{N} - \frac{\hat{x}^{(m)} \hat{x}^{(i)}}{N}) = \frac{a}{s} \left[g^{(i)} - \frac{1}{N} \sum_{m=1}^{N} g^{(m)} - \frac{\hat{x}^{(i)}}{N} \sum_{m=1}^{N} g^{(m)} \hat{x}^{(m)} \right]$